

Training on Foundation of Programming Skills - III - Data Structures & Algorithms

Sl. NO	Modules	Topics	Hands-on Practice
1	Introduction to DSA	What is Data Structure & Algorithm?	Write 3 real-life examples (array = list of marks, stack = undo feature)
		Types of Data Structures (Linear & Non-Linear)	Classify 10 structures into linear/non-linear
		Time & Space Complexity	Calculate complexity of simple loops
		Big-O, Big-Theta, Big-Omega	Analyze Best, Average, Worst Case
		Best, Average, Worst Case Analysis	Compare linear search cases
		Algorithm Design Basics	Write steps for finding third max in array
		Contest-1 On Basics	
		Introduction to Arrays	Create array + print elements
		Operations (Insertion, Deletion, Traversal)	Implement manually (no built-in shortcuts)
		Searching (Linear, Binary Search)	Code both + compare time
		Sorting (Bubble, Selection, Insertion)	Implement all 3
2	Arrays	Two Pointer Technique	Solve sums using two pointer
		Sliding Window Technique	Solve Max sum subarray of size k
		Prefix Sum Technique	Solve Range sum queries
		Kadane's Algorithm	Solve Maximum subarray sum
		Binary Search on Answer (Very Important)	Learn different methods to solve and try it in leetcode under binary search tab
		Array Problem Solving	Practice 10 mixed problems
		Contest-2 On Arrays	
		Introduction to Strings	Traverse string
		String Manipulation	Reverse string
		Pattern Matching (Naive)	Find substring manually
3	Strings	KMP Algorithm (pattern matching)	Implement LPS array
		Anagram, Palindrome Problems	Check both
		String Problem Solving	Solve 5-10 problems
		Contest-3 On Strings	

		Basics of Recursion		Try Solving Basic recursion problems (print 10 numbers)
		Recursive Problem Solving	Solve Factorial, Fibonacci	
4	Recursion & Backtracking	Backtracking Concept	Generate subsets	
		Problems (N-Queens, Subsets, Permutations)	Solve at least 1 each	
		Contest-4 On Recursion	Create node class	
		Introduction to Linked List	Insert at beginning/end	
		Singly Linked List	Add prev pointer	
		Doubly Linked List	Connect last to first	
		Circular Linked List		
		Operations (Insert, Delete, Traverse)	Implement all	
		Fast & Slow Pointer (Cycle detection)	Detect cycle	
		Linked List Problems	Solve 5 linked list medium level problems	
5	Linked List	Contest-5 On Linked List	Implement using array	
		Introduction to Stack	Code both	
		Implementation (Array & Linked List)	Try and Test all	
		Operations (Push, Pop, Peek)	Solve valid parentheses	
		Applications (Expression Evaluation, Parenthesis Matching)	Solve Next greater element	
		Monotonic Stack (Interview Important)	Solve 5 problems	
		Stack Problems	Implement queue	
		Contest-6 On Stack	Code circular queue	
		Introduction to Queue	Practice both	
		Types (Simple, Circular, Priority, Deque)	Enqueue, Dequeue	
7	Queue	Implementation (Array & Linked List)	Solve First non-repeating character	
		Queue Operations		
		Queue Problems		
		Contest-7 On Queue		

8	Trees	Introduction to Trees	Create tree node
		Binary Tree	Insert nodes
		Binary Search Tree (BST)	Insert & search
		Tree Traversals (Inorder, Preorder, Postorder)	Code Inorder, Preorder, Postorder
		Height, Depth, Levels	Calculate height of a Tree
		Tree Problems	Solve Lowest Common Ancestor
		Contest-8 On Trees	
		Introduction to Heap	Understand min/max heap
		Min Heap & Max Heap	Insert & delete
		Heap Operations	Understand Heapify and code
		Priority Queue	Use built-in PQ
9	Heap	Heap Problems (Top K, Heap Sort)	Solve Top K frequent elements
		Contest-9 On Heap	
		Introduction to Hashing	Use hashmap, hashset, linked hashmap and linked hashset
		Hash Functions	Understand mapping
		Collision Handling Techniques	Learn chaining
10	Hashing	Hash Tables	Implement basic and Solve basic Hash table problems
		Hashing Problems	Solve problems related to frequency count.
		Contest-10 On Hashing	
		Introduction to Graphs	Learn about structure and types of graphs
		Graph Representation	Represent graph using matrix and adjacency list
		BFS (Breadth First Search)	Traverse graph
		DFS (Depth First Search)	Traverse recursively
		Topological Sort	Solve DAG ordering
		Shortest Path Algorithms (Dijkstra)	Implement basic login and Solve minimum path sums in graph
		Disjoint Sets	Solve problems based on Union-Find
11	Graphs	Graph Problems	Solve Problems like Number of Islands and Rotten Oranges
		Contest-11 On Graphs	

12	Sorting & Searching (Advanced)	Merge Sort	Implement Merge Sort
		Quick Sort	Implement Quick Sort
		Binary Search Advanced	Search in rotated array
		Divide & Conquer Technique	Apply in merge sort
		Sorting Problems	Solve 5 problems
		Contest-12 On Sorting	
13	Greedy Algorithms	Introduction to Greedy	Understand greedy choice
		Activity Selection	Solve scheduling
		Fractional Knapsack	Solve Maximize profit and Related Problems
		Huffman Coding	Build tree using Huffman
		Greedy Problems	Solve 5 problems
		Contest-13 On Greedy	
14	Dynamic Programming	Introduction to DP	Solve Fibonacci using DP
		Memoization & Tabulation	Convert recursion to DP
		Fibonacci, Knapsack	Solve both related problems in Leet code
		DP on Grids(Path Problems)	Solve Unique paths problems
		Longest Common Subsequence	Implement and Solve subsequence problems in leetcode
		DP on Trees(Basics)	Try Basic tree DP Implementation
		DP Problems	Solve 5-10 problems
		Contest-14 On DP	